EE Assignment 3

C–V Characteristics of a MOSCAP

M. Kalanidhi Maran, Roll No: EE24B028

Sri Gayathri V, Roll No: EE26E008

Shaun Joseph, Roll No: EE26E007

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1 Introduction

This report presents the simulated high-frequency (HF) and low-frequency (LF) C–V characteristics of a MOS capacitor (MOSCAP), obtained using a Verilog-A quasi-static charge model implemented in the ADS/Spectre-like simulator. The gate–body capacitance was extracted as $C = \dfrac{I_d}{2\pi f\, V_{ac}}$, where the AC excitation frequency f was kept constant 1 Hz (LF) and 1 kHz (HF).

Note on convergence: at $V_{gb} = 0$ the Newton–Raphson solver for the surface potential ϕ_S can fail to converge (division by $\phi_S \to 0$ in the error metric). To avoid this, V_{gb} was taken to be very small value at which capacitance was measured and debye length was measured from it.

2 C–V Curves: Variation with Substrate Doping N_a

N_a was varied from $1 \times 10^{17}\,\mathrm{cm}^{-3}$ to $5 \times 10^{18}\,\mathrm{cm}^{-3}$ (4 intermediate steps shown), with t_{ox} held fixed at 5 nm.

2.1 Low-Frequency C–V vs N_a

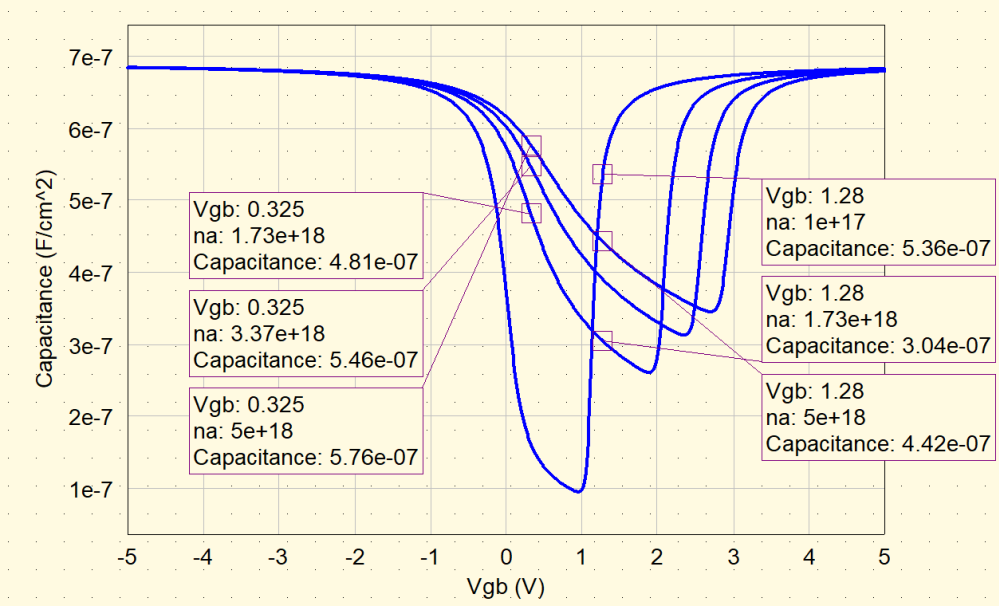


Figure 1: Low-frequency C–V curves for $N_a = 1 \times 10^{17}$ to $5 \times 10^{18}\,\mathrm{cm}^{-3}$ (4 curves total), $t_{ox} = 5\,\mathrm{nm}$.

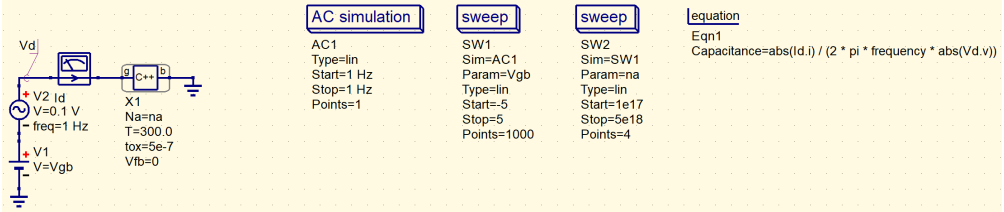


Figure 2: Simulation Schematic

As N_a increases, the accumulation capacitance barely changes (~ 5.7–5.8×10^{-7} F/cm^2), since it depends only on the oxide thickness and not on doping. However, the inversion-side capacitance rises noticeably with doping, because higher doping results in a thinner maximum depletion width, pushing the total capacitance closer to C_{ox}. The dip in the curve also becomes narrower and shifts toward higher $|V_{gb}|$ at higher N_a, since heavily doped substrates require more applied voltage to reach inversion, and the depletion region is thinner overall.

2.2 High-Frequency C–V vs N_a

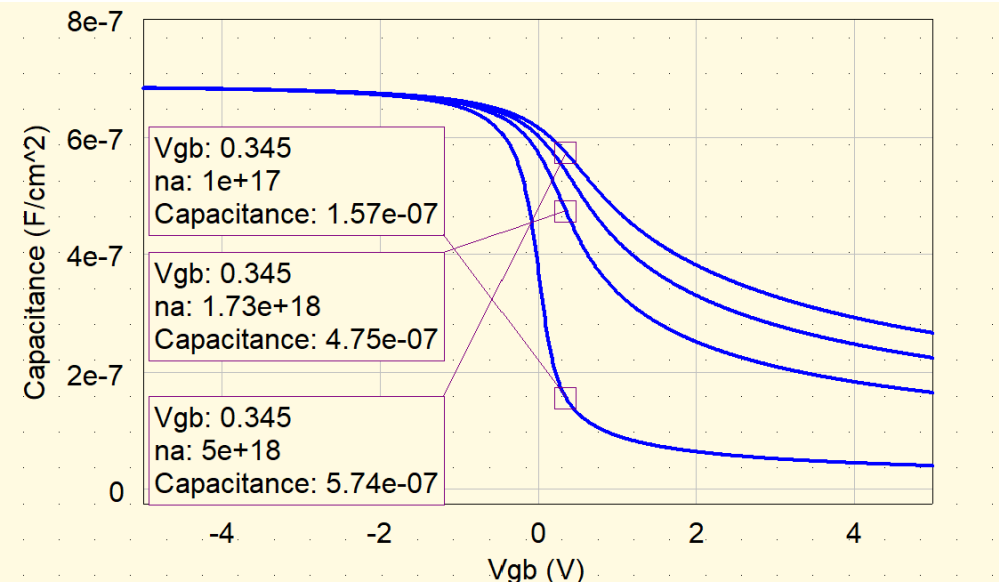


Figure 3: High-frequency C–V curves for $N_a = 1 \times 10^{17}$ to $5 \times 10^{18}\,\mathrm{cm}^{-3}$ (4 curves total), $t_{ox} = 5\,\mathrm{nm}$.

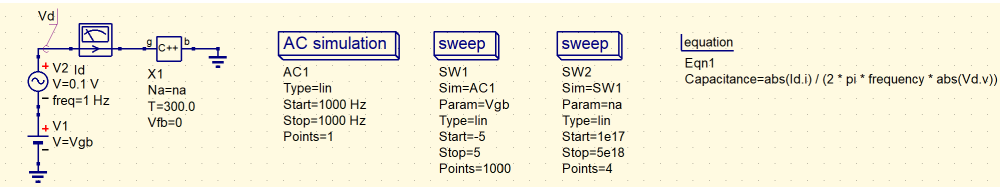


Figure 4: Simulation Schematic

At high frequency, the accumulation-side capacitance converges to roughly the same value ($\sim$ 5.7–6 $\times$ 10^{-7} F/cm^2) across all doping levels, since it depends only on C_{ox}. Unlike the LF case, the inversion-side capacitance stays permanently low ($\sim$ 1.5–2 $\times$ 10^{-7} F/cm^2) instead of recovering, since minority carriers can't respond fast enough to the AC signal. The capacitance is instead set by the maximum depletion width in series with C_{ox}: $C_{min} = C_{ox}C_{dmax}/(C_{ox} + C_{dmax})$. As N_a increases, W_{max} shrinks, so C_{dmax} rises and C_{min} increases with doping, seen here as the minimum going from $\sim 1.57 \times 10^{-7}$ at $N_a = 1 \times 10^{17}$ to higher values at $N_a = 5 \times 10^{18}$ cm^{-3}. The point, at which curve falattens towards the right, also shifts toward larger $|V_{gb}|$ with increasing N_a.

3 C–V Curves: Variation with Oxide Thickness t_{ox}

t_{ox} was varied from 5 nm to 15 nm (4 intermediate steps shown), with N_a held fixed at 1×10^{17} cm^{-3}.

3.1 Low-Frequency C–V vs t_{ox}

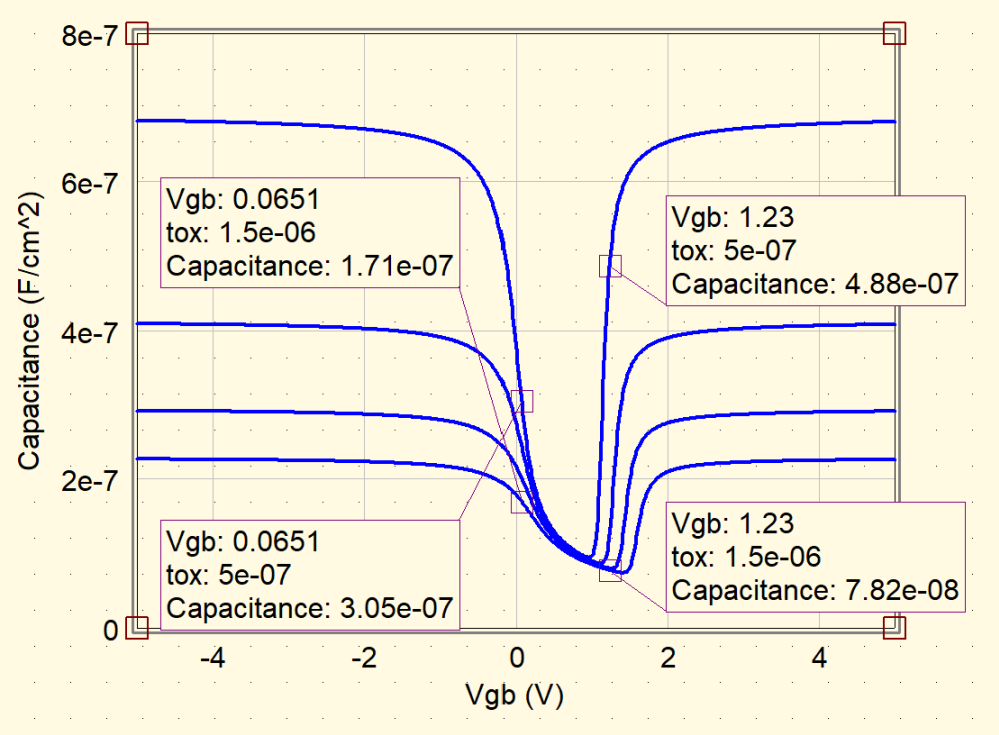


Figure 5: Low-frequency C–V curves for t_{ox} = 5 to 15 nm (4 curves total), $N_a = 1 \times 10^{17}$ cm^{-3}.

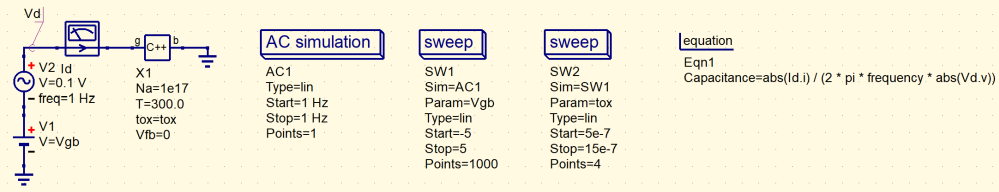


Figure 6: Simulation Schematic

As t_{ox} increases from 5 to 15 nm, both the accumulation and inversion capacitances drop together, since $C_{ox} = \varepsilon_{ox}/t_{ox}$ sets the overall scale. At $t_{ox} = 5\,\text{nm}$, the value is $\sim 4.88 \times 10^{-7}$ F/cm^2, falling to $\sim 1.71 \times 10^{-7}$ F/cm^2 at $t_{ox} = 15\,\text{nm}$ – roughly a threefold drop matching the threefold increase in t_{ox}. Since N_a is fixed, C_{dmax} stays the same, but the smaller C_{ox} for thicker oxides pulls the whole curve down uniformly, without changing the shape or position of the dip.

3.2 High-Frequency C–V vs t_{ox}

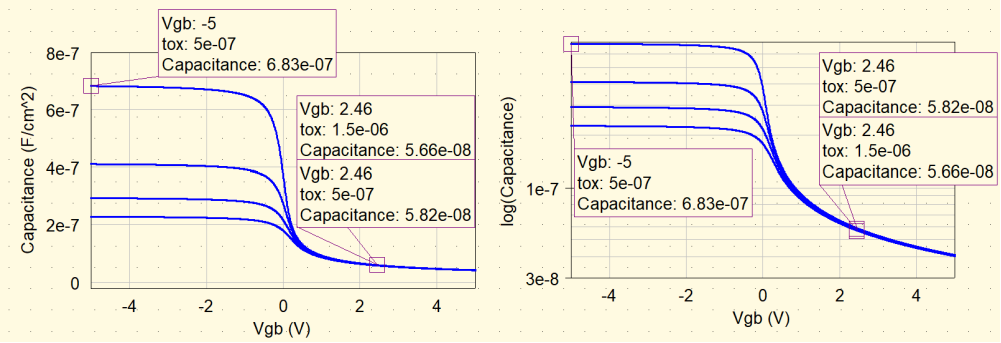


Figure 7: High-frequency C–V curves for $t_{ox} = 5$ to 15 nm (4 curves total), $N_a = 1 \times 10^{17}\,\text{cm}^{-3}$.

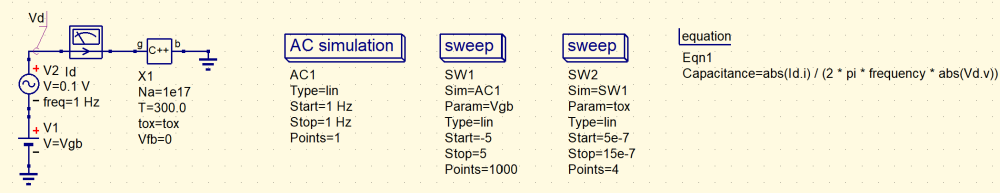


Figure 8: Simulation Schematic

As t_{ox} increases from 5 to 15 nm, the accumulation-side capacitance again scales down with $C_{ox} = \varepsilon_{ox}/t_{ox}$, dropping from $\sim 6.83 \times 10^{-7}$ F/cm^2 at $t_{ox} = 5\,\text{nm}$ to $\sim 2.28 \times 10^{-7}$ F/cm^2 at $t_{ox} = 15\,\text{nm}$. On the inversion side, the capacitance stays pinned at its low HF minimum as minority carriers still can't respond. The log-scale plot confirms this trend clearly: thicker oxide compresses both the accumulation plateau and the inversion floor toward smaller values, while N_a being fixed keeps the depletion width – and hence the shape and position of the transition unchanged.

4 Debye Length Calculation

Theoretical Debye Length

For $N_a = 1 \times 10^{17}$ cm^{-3} and $t_{ox} = 5\,\text{nm}$, using $\varepsilon_{si} = 11.9\,\varepsilon_0 = 1.0536 \times 10^{-10}$ F/m, $k_B = 1.380649 \times 10^{-23}$ J/K, $T = 300\,\text{K}$, and $q = 1.602 \times 10^{-19}$ C:

$$L_D = \sqrt{\frac{\varepsilon_{si} k_B T}{q^2 N_a}} \tag{1}$$

$$L_D = \sqrt{\frac{(1.0536 \times 10^{-10})(1.380649 \times 10^{-23})(300)}{(1.602 \times 10^{-19})^2 (1 \times 10^{23})}} \tag{2}$$

$$L_D \approx 13.04 \text{ nm} \tag{3}$$

Simulated Debye Length

Since $V_{fb} = 0$ is set in the Qucs Studio simulation, the flat-band capacitance C_{FB} is read directly at $V_{gb} = 0$ from the simulated C–V curve, giving $C_{FB} = 3.2 \times 10^{-7}$ F/cm^2. The oxide capacitance is

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.9\,\varepsilon_0}{5\,\text{nm}} \approx 6.9 \times 10^{-7} \text{ F/cm}^2.$$

The semiconductor capacitance at flat-band is extracted from the series combination:

$$\frac{1}{C_s} = \frac{1}{C_{FB}} - \frac{1}{C_{ox}} \tag{4}$$

$$\frac{1}{C_s} = \frac{1}{3.2 \times 10^{-7}} - \frac{1}{6.9 \times 10^{-7}} = 1.676 \times 10^6 \text{ cm}^2/\text{F} \tag{5}$$

$$C_s \approx 5.97 \times 10^{-7} \text{ F/cm}^2 \tag{6}$$

The extracted (simulated) Debye length is then

$$L_D^{sim} = \frac{\varepsilon_{si}}{C_s} \tag{7}$$

$$L_D^{sim} \approx 17.7 \text{ nm} \tag{8}$$

Comparison

	Theoretical	Simulated
L_D	13.04 nm	17.7 nm

The simulated Debye length deviates from the theoretical value by approximately 35%, which is reasonably close given the sensitivity of reading C_{FB} precisely at $V_{gb} = 0$ from the steep depletion-region slope of the C–V curve.

5 High-Frequency C–V with $V_{FB} = -1\,\mathrm{V}$

Here the flat-band voltage was shifted to $-1\,\mathrm{V}$ (e.g. via a work-function difference / fixed oxide charge term added to the model), and the HF C–V curve was re-simulated.

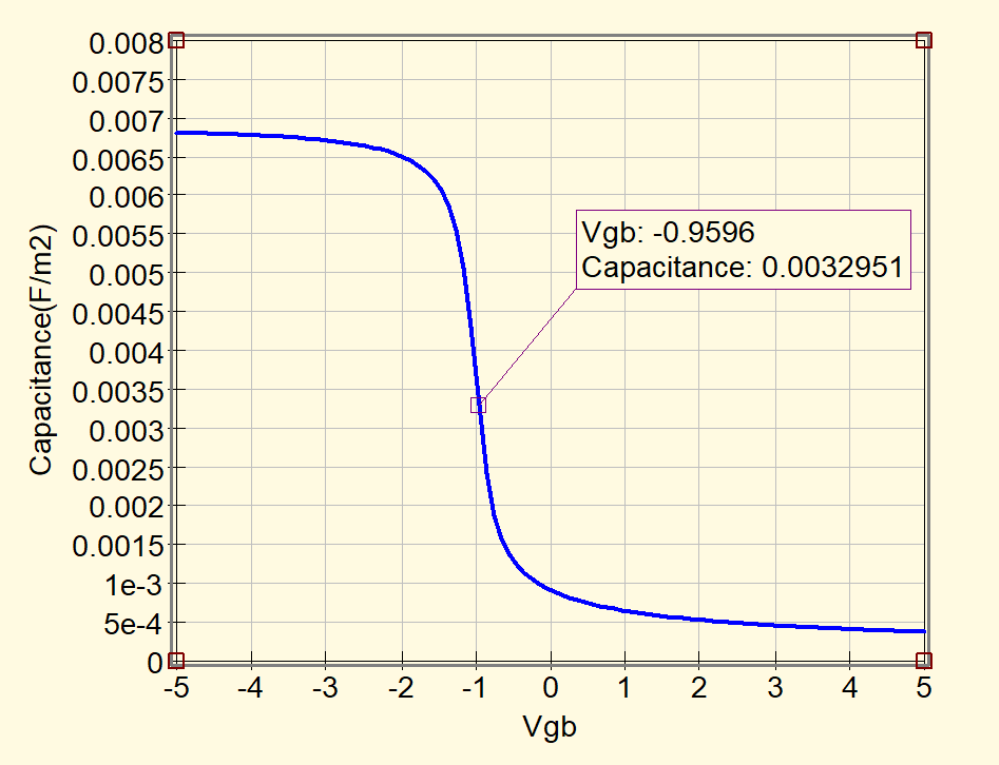


Figure 9: High-frequency C–V curve with $V_{FB} = -1\,\mathrm{V}$.

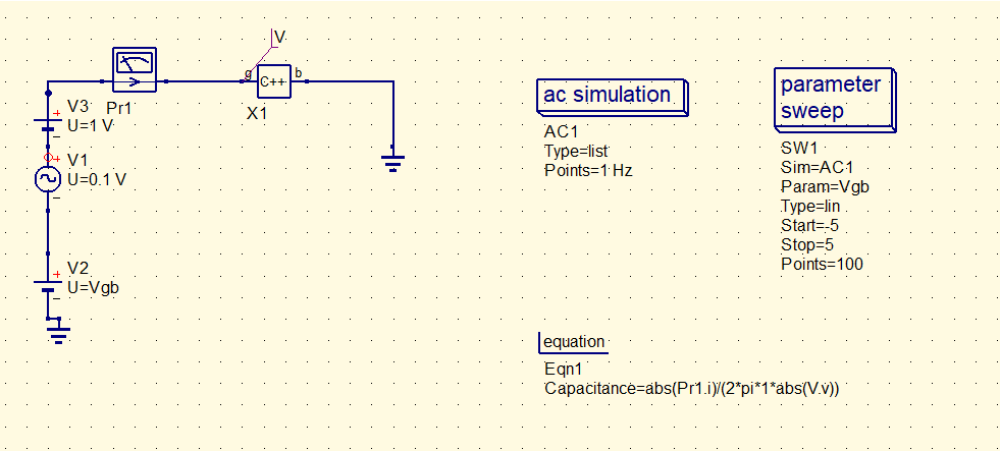


Figure 10: Simulation Schematic

Introducing $V_{FB} = -1\,\mathrm{V}$ rigidly shifts the entire HF C–V curve along the voltage axis by $-1\,\mathrm{V}$, with no change in its shape. This is expected physically, since V_{FB} only redefines the

reference point at which band bending is zero ($\phi_S = 0$ occurs at $V_{gb} = V_{FB}$ rather than at $V_{gb} = 0$), while the accumulation-side and inversion-side (HF minimum) capacitance remain unchanged. The shift therefore confirms that the model correctly separates the "where" of the curve (set by V_{FB}, i.e. gate work function or fixed oxide charge) from its "shape" (set by the semiconductor doping and oxide capacitance).

6 Note on Numerical Convergence Near $V_{gb} = 0$

At $V_{gb} = 0$, the Newton–Raphson solver for the surface potential ϕ_S was found to fail to converge.This is a direct consequence of the error metric used in the iteration,

$$\text{error} = \frac{\phi_{S,\text{new}} - \phi_S}{\phi_S},$$

which normalizes the update step by ϕ_S itself. Since $\phi_S \to 0$ exactly at flat-band (by definition, $V_{gb} = V_{FB}$ corresponds to zero band bending), this relative error metric blows up (division by a value approaching zero), causing the loop to either never satisfy the convergence tolerance or to produce `NaN`/`inf` results despite ϕ_S itself having essentially converged.

This was resolved by sweeping V_{gb} from a small nonzero offset instead of exactly 0 V (e.g. $V_{gb} = \pm 10^{-3}$ V), which kept ϕ_S bounded away from zero and allowed the relative error metric to remain well-defined throughout the iteration. A more robust long-term fix would be to switch to an *absolute* error metric, error $= \phi_{S,\text{new}} - \phi_S$, which does not depend on the magnitude of ϕ_S and therefore remains numerically stable even exactly at flat-band.

7 Conclusion

This assignment verified MOSCAP behavior through simulated C–V characteristics. Increasing N_a raised the inversion/minimum capacitance and narrowed the depletion dip, while increasing t_{ox} uniformly scaled down the curve via $C_{ox} = \varepsilon_{ox}/t_{ox}$, with the shape governed independently by doping and oxide thickness. The HF curve's failure to recover in inversion, unlike LF, reflects the inability of minority carriers to follow a fast AC signal. The extracted Debye length (~ 17.7 nm) reasonably matched theory (~ 13.0 nm), and shifting V_{FB} to -1 V confirmed it simply translates the curve without changing its shape. Overall, the simulations closely reproduced standard MOSCAP theory.

8 Verilog A Code

Please refer below the code used for high frequency simulation including the effect of flat-band voltage.

Listing 1: Verilog-A model for MOSCAP

```
1 `include "disciplines.vams"
2 `include "constants.vams"
3 module moscap(g,b);
```

```
4       inout g,b;
5       electrical g,b;
6
7       parameter real Na = 1e17 from (0:inf); // Bulk doping
           concentration (cm^-3) (p=type)
8       parameter real T = 300.0; // Absolute Temperature (K)
9       parameter real tox = 5e-7; // Oxide thickness (cm)
10      parameter real tolerance = 1e-10;
11      parameter real Vfb = -1;
12      parameter real Eg = 1.12; //Bandgap in eV
13      real eox, es, phiB, ni,Vt,Vth, Cox, gamma, psiS0, term;
14      real Vgb, psiS, errorFraction, error, smoothSgn, smoothSgnDerive
           , A1, A2, A3, A1Derive, A2Derive, A3Derive, Qs, fPsiS,
           fPsiSDerive, Q0, psiSNew, Qg, delta;
15      real Igb;
16      real Nc300, Nv300; //eff. DOS, valence band @300K in cm^-3
17      real NcT, NvT;
18
19      analog begin
20          Vgb=V(g,b);
21      // -------------------------Constants
           ----------------------------------
22          eox = 1e-2 * 3.9 * `P_EPS0; //converting to Fcm^-2
23          es   = 1e-2 * 11.7 * `P_EPS0; //converting to Fcm^-2
24          Cox = eox/tox; //(Fcm^-2)
25          ni = 1e10; // Intrinsic Carrier Concentration
26          Vt = `P_K*T/`P_Q;// Volt
27          phiB = Vt*ln(Na/ni); // Volt
28          gamma = sqrt(2.0*es*`P_Q*Na)/Cox;
29          Vth = 2.0*phiB + gamma*sqrt(2.0*phiB) + Vfb;
30          term = (Vgb-Vfb-2.0*phiB)/(gamma*sqrt(Vt));
31          Nv300 = 1.02e19;
32          Nc300 = 2.8e19;
33          NcT = Nc300*pow((T/300), 1.5);
34          NvT = Nv300*pow((T/300), 1.5);
35
36          // Modelling intrinsic carrier concentration
37          ni = sqrt(NcT*NvT)*limexp(-Eg*(`P_Q)/(2*(`P_K)*T));
38      // -------------------------------------------------------------
39
40      // Seed for Newton-Ralphson
41          if (Vgb-Vfb>=0.0) begin
42              if (Vgb<Vth) begin
43              psiS0 = Vgb - Vfb - ((gamma*gamma/2.0)*(sqrt(1.0+(4.0*(
                   Vgb-Vfb)/(gamma*gamma)))-1.0));
44          end else if (Vgb>=Vth) begin
45              psiS0 = 2.0*phiB + Vt*ln(term*term-(2.0*phiB/Vt)+1.0);
```

```
46          end
47          end
48          else begin
49              psiS0 = -0.1;
50          end
51
52      // Running Newton-Ralphson for getting psiS for given Vgb
53          errorFraction = 1.0;
54          error = 1.0;
55          delta = 1e-10;
56          psiS = psiS0;
57          while (abs(error)>tolerance) begin
58              smoothSgn = psiS/sqrt(psiS*psiS+delta);
59              A1 = limexp(-1.0*psiS/Vt)-1.0;
60              A1Derive = (-1.0)*limexp(-1.0*psiS/Vt)/Vt;
61              A2 = (psiS/Vt)*(1.0-limexp(-2.0*phiB/Vt));
62              A2Derive = (1.0-limexp(-2.0*phiB/Vt))/Vt;
63              A3 = 0;
64              A3Derive = 0;
65              Q0 = gamma*sqrt(Vt)*Cox;
66              Qs = (-1.0)*Q0*smoothSgn*sqrt(A1+A2+A3);
67              fPsiS = psiS - (Qs/Cox) - Vgb + Vfb;
68              smoothSgnDerive = delta / pow(psiS*psiS + delta, 1.5);
69              fPsiSDerive = 1.0 + (Q0/Cox)*((smoothSgn)*(A1Derive +
                    A2Derive + A3Derive)/(2.0*sqrt(A1+A2+A3)) + sqrt(A1+
                    A2+A3)*smoothSgnDerive ); // WE ARE NOT
70              // INCLUDING THE DERIVATIVE OF THE SMOOTH SGN FN IN THE
                    GIVEN CODE. BUT HERE I HAVE INCLUDED
71
72              psiSNew = psiS - (fPsiS/fPsiSDerive);
73              //errorFraction = (psiSNew-psiS)/psiS; // IF VGB = 0,
                    PSI = 0, WHICH MEANS THE ERROR TERM BLOWS UP
74              error = (psiSNew-psiS);
75              psiS = psiSNew;
76          end
77
78          Qg = Cox * (Vgb-Vfb-psiS);
79          I(g,b) <+ ddt(Qg);
80          $strobe("Na=␣%g,␣Vfb=␣%g,␣Vth=␣%g,␣error=␣%g.\n",Na,Vfb, Vth
                , error);
81      end
82  endmodule
```